Gerald Emanuel Vargas Morales

Computer Engineering Student · Full Stack Developer · Data Analysis · Applied AI

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PROFILE

Final-year Computer Engineering student at **Universidad Mayor de San Simón (UMSS)** with solid technical training and **3+ years of experience delivering paid projects** for businesses and institutions. Contributed to the development of **electoral systems** for the Cochabamba Sub-national Elections, mobile applications for the Municipal Government of Colcapirhua, management systems for local businesses, and modern educational platforms. Experienced in **deploying and maintaining multiple production systems on Linux servers**. Committed to technical excellence and continuous learning, with a strong focus on **Artificial Intelligence** and **Machine Learning**.

PROFESSIONAL EXPERIENCE

Full Stack Developer — Official Internship

January 2026 – April 2026

Gobierno Autónomo Municipal de Colcapirhua · Laravel 11 + Flutter 3.41

- Developed **ColcaTrufis**: cross-platform urban mobility app (Trufi/Radiotaxi) with official municipal route data.
- Stack: **Flutter** (mobile frontend) + **Laravel** (RESTful APIs). Development and maintenance of institutional websites.
- Digital solutions for optimising municipal administrative processes.

Full Stack Developer — Electoral Ballot System (NGP Party)

December 2025 – March 2026

Sub-national Elections Cochabamba · Node.js + React **CURRICULAR CERTIFICATE**

- **Successfully deployed** and used during the official Cochabamba sub-national electoral process.
- Backend: **Node.js/Express** — RESTful APIs for real-time electoral data processing.
- Frontend: **React** — interactive dashboard for live visualisation of results. Scalable clean architecture.

Full Stack Developer — Inventory & Management System

2026 · In Production Use

Ferretería Urkupiña · Laravel 12 · React · Tailwind CSS **PAID PROJECT**

- Designed and developed a complete **inventory and product management system** for a hardware store, currently **in active use** by the business.
- Backend: **Laravel 12**. Frontend: **React + Tailwind CSS** — custom dashboard with interactive stats cards, product tracking and stock control.
- Focused on clean UI/UX with custom animations and a cohesive brand identity for day-to-day operational use.

Full Stack Mobile Developer — Electricity Meter Reading System

2024

Cooperativa de Electricidad "15 de Noviembre" — Potosí, Bolivia · Flutter · Laravel · ESC/POS · Bluetooth · OpenStreetMap **PAID PROJECT**

Android mobile application developed for field technicians of the **"15 de Noviembre"** electrical cooperative (Potosí, Bolivia). Enabled on-site meter reading registration with GPS geolocation and automatic thermal receipt printing via Bluetooth. Backend built with **Laravel**, consuming the cooperative's existing system REST APIs. Full management of Bluetooth peripheral lifecycle and runtime Android permissions.

- Field technician authentication and meter reading registration with GPS coordinates.
- Automatic pre-notice printing via **Bluetooth ESC/POS** thermal printer upon each reading.
- Full thermal printer management: device scanning, pairing, and automatic reconnection.
- Interactive client map rendered with **flutter_map + OpenStreetMap** (no API key required).
- REST API integration with the cooperative's legacy system; local printer config persistence.

Process Automation with n8n

March 2026

Automation Projects (in collaboration)

- Automated bulk migration of **6,500+ Excel rows** into the ballot system with full traceability.

- Multi-source data integration and validation via **n8n** workflow orchestration.

Full Stack Developer — Church Information Website (IENAD BETANIA RESTAURACION) *2026 · In Production Use*
Personal Project · Python · Streamlit

- Built and deployed a website for a local church, currently **in active use**, providing up-to-date information on church events.
- Includes links to the church's social media and relevant information about the congregation and its activities.
- Developed entirely with **Python + Streamlit** as a lightweight, easy-to-maintain solution.

FEATURED PROJECTS

Centinela — Hackathon By Paseo Aranjuez *October 2026*
Python · FastAPI · Next.js · Docker **HACKATHON WINNER**

- Took part in the **By Paseo Aranjuez** hackathon and **won the remote-work prize** for the company **On Business (Colombia)** with the project **Centinela**.
- Backend built with **Python + FastAPI**, developed as a team project.

Synapse — Comprehensive Educational Platform *2025 – 2026*
Laravel 12 · PHP 8 · MySQL · JavaScript

- Full-featured LMS: course management, user administration and academic content delivery.
- Role-based access (Student / Teacher / Admin) with personalised dashboards per profile.
- Automated assessments, progress tracking, and detailed admin reports. Optimised with caching and eager loading.

Financial Fraud Detection with Deep Learning *2025 – 2026*
Python · Streamlit · TensorFlow/Keras · Scikit-learn · Pandas · Plotly

- Hybrid **Isolation Forest + Autoencoder** system (20-12-8-12-20 architecture) for anomaly detection.
- Pipeline for imbalanced data (fraud <0.1%): **94.2% precision / 91.8% recall** on 150,000+ transactions.
- Interactive Streamlit dashboard with anomaly visualisation and adaptive risk scoring.

TECHNICAL SKILLS

Backend	Python, FastAPI, PHP, Laravel, Node.js, Express.js, Java, C++
Frontend	React, Next.js, Astro, Flutter, HTML5, CSS3, JavaScript
Data / ML	Pandas, NumPy, TensorFlow, Keras, Scikit-learn, Streamlit
Databases	MySQL, PostgreSQL, MongoDB, SQLite, SQL Server
Tools	Git, GitHub, n8n, Docker, Linux, Jira

EDUCATION

Bachelor of Computer Engineering *2024 – Present*
Universidad Mayor de San Simón (UMSS) — Cochabamba, Bolivia

CERTIFICATIONS & COURSES

- Intensive Python (NumPy, Pandas)
- Machine Learning (Microsoft)
- Recursion and Backtracking
- Competitive Programming — ICPC-UMSS
- Web Development with Laravel

ACHIEVEMENTS

✓ **Hackathon Winner — By Paseo Aranjuez:** Won the **remote-work prize** for On Business (Colombia) with **Centinela** (Python + FastAPI).

✓ **Curricular Certificate:** Official Electoral Ballot System — Sub-national Elections Cochabamba 2026.

✓ **ICPC Bolivia 2026 (two phases):** Ranked **#9** in Phase 1 and **#10** in Phase 2 at Cochabamba level; ranked **#20** nationwide (Bolivia) with my team.

✓ **Production Deployment:** ColcaTrufis now available on the Google Play Store under the name **ColcaTrufis**.

✓ **Successful Electoral System:** Actively used in the official Cochabamba 2026 electoral process.

✓ **Production Deployment:** Inventory system for Ferretería Urkupiña in active daily use.

✓ **Server Deployment:** Deployed and maintained multiple production systems on **Linux servers**.

LANGUAGES

Spanish	Native
English	B1